

# Feeding the fish: Interaction design to support listening to accounts of marginalization

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## ABSTRACT

We present a study using an immersive, virtual reality experience designed to deepen understandings of gender and sexuality-based marginalization in Science, Technology, Engineering, and Math (STEM) as complex experiences with individual and systemic dimensions. A challenge to teaching about marginalization is that learners might disengage from the unpleasant experience of listening to first-hand accounts of discrimination. A further challenge is encouraging learners to acknowledge their role in producing those harms. Through our design of playful, embodied interactions in virtual reality, the app effectively supports learners to remain engaged in listening to and learning about the marginalized other. As part of a design study, the first author interviewed participants over Zoom as they interacted with the VR game. In our analysis, we describe our interaction designs and connect these to analysis showing how participants reflect upon and use the interactions to move through the difficulties of listening to another's stories of marginalization.

**Index Terms:** Human-centered computing—Empirical studies in interaction design; Human-centered computing—Virtual reality; Applied computing—Interactive learning environments

## 1 INTRODUCTION

We present Mementorium, an interactive, branching narrative told in immersive, virtual reality (VR). The player uncovers the narrator's memories of gender and sexuality-based marginalizations in STEM learning environments, moving from childhood to early adulthood. Mementorium's design builds upon our previous designs and research on queer reorientations to computing and queer approaches to embodied learning in VR [3, 5]. The design also draws upon our personal experiences, research on LGBTQ+ exclusion from STEM disciplines, and the first author's 10+ years of experience as a DEI practitioner. Our design aims to increase understanding of interpersonal and societal structures contributing to women's and LGBTQ+ exclusion from STEM learning and professions. Our aims are accomplished through the interaction design, which encourages players to remain oriented to the narrator's experience and to reflect upon their experience.

Teaching about marginalization can be challenging as learners might disengage from the unpleasant experience of listening to first-hand accounts of discrimination or resist reflecting upon the systems that engender their marginalization or that they may be implicated in reproducing.

Media scholar Lisa Nakamura [2] calls VR documentaries for social good *Virtuous VR*, and argues that this new media, much like earlier forms of media such as literature, photography, and film, “use representations of suffering to produce feelings of immersion, identification, and empathy for those bodies that are not ours” (p. 53).

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Nakamura [2] argues that from viewers' perspectives, experiencing the pain of the marginalized other produces a sense of virtue without guilt. *Virtuous VR* does not expect players to trust marginalized people's accounts of their suffering. Instead, it puts them into a marginalized body to experience the suffering and make their own judgments.

Conversely, Mementorium requires players to dialogue with marginalized people and consider the interpersonal actions and societal structures perpetuating harm and exclusion. Players must reflect upon and choose counter-actions that challenge exclusionary experiences.

Using Mementorium, we investigate how virtual reality can be leveraged to support productive and playful inquiry about gender and sexuality. We describe two of our interaction designs and connect these to analyses showing how participants reflect upon and use the interactions to move through the difficulties of listening to another's stories of marginalization.

## 2 METHODS

### 2.1 VR interaction design

Mementorium was developed in Unreal Engine 4.25 (Unreal) [1].

#### 2.1.1 Feed The Fish

In the Feed the Fish interaction, the player discovers a school of vibrantly colored fish swimming through the air in the game's “magical realm.” Behind the fish, there is a can of magical fish food, and when the player turns it upside down and shakes it, fish food flakes come out and attract the fish towards the player (Figure 1a). Fish food cans appear under each story choice, allowing the player to choose the direction of the narrative by feeding the fish. The movement of the fish is controlled by Reynolds' [9] flocking algorithm, which gives them a real-time, dynamic ability to interact with and “flock” towards the player.

#### 2.1.2 Code Poetry

In the Code Poetry interaction, the player is introduced to the narrator's experiences of a high school coding classroom. Next, the player is presented with a block-based coding interaction where the blocks they choose to complete the code also reflect their branching story choices. As the code is entered, the player constructs Python-based code poetry that reflects experiences of gender, sexuality, and belonging in coding learning environments (Figure 1b).

### 2.2 Data Collection

As part of a design study, the first author interviewed participants as they interacted with the VR game. The study setting was over Zoom videoconferencing, in which participants were provided access to the VR application and played the game while screen sharing their gameplay screen. Participants included people of various genders and sexualities, ranging from their early 20s to mid-50s, and geographic locations including Canada, the United States, Australia, and Switzerland. The study involved a single session, approximately 2 hours, including a pre-interview, the interaction with the VR game, and a post-interview.

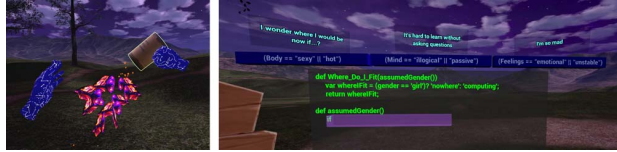


Figure 1: Screenshot of the VR Game, Mementorium, displaying (a, right) the feed the fish interaction (b, left) the poetry coding interaction.

### 3 RESULTS

We provide two illustrative examples to show how participants' interactions supported their emotional connection to the narrator and reflection upon their experiences in relation to gender and sexuality-based systems of marginalization.

#### 3.1 Illustrative Case 1: Feed the Fish

Participants hear the narrator's story about going to a postsecondary open house during the feed the fish interaction. The narrator is excited about the data science and computer modelling program booth but is dismissed by the man running the booth. One participant explained that the effect of the fish interaction is therapeutic, supporting her listening to the narrator's experience of marginalization:

Casey: I think even like it's a very positive interaction in that I'm giving something to these animals. They're following me like it has a real social kind of relational way of doing things which felt very in contrast to the story of the guy being an asshole. So it had a real, like almost therapeutic opposite interaction.

#### 3.2 Illustrative Case 2: Code Poetry

Participants co-constructed the meaning of the code poetry to tell both the narrator's story about what gender and sexuality in a computing learning environment were like for the narrator and what gender and sexuality have meant to the participants personally. For example, Clare, a nonbinary person, reflected closely on both the narrative choices and the code poetry blocks:

Clare: [Reads a narrative choice aloud "It's hard to learn without asking questions."] I had a hard time asking questions in class. I still do. Because I had some bad experiences with some teachers when I was young. When I was young, I had a lot of anger. In my case, it was probably because as a "boy," I wasn't allowed to have tender emotions. Like if I cried, people would make fun of me or my parents would like fucking shame me. But being angry was more acceptable. Also, all this fucking bullshit made me really upset. And then I would get angry. I'm going to choose this one because I've always wondered what it would be like if I had a different body. [Chooses the story option "I wonder where I would be now if...?" with the block code stating, "(body == "sexy" || "hot")"].

### 4 DISCUSSION

Our illustrative cases show two different but complementary effects of the interaction design. One player, Casey, was supported to continue listening to the narrator's experience of marginalization because the feed the fish interaction provides a therapeutic, opposite effect to the story. Meanwhile, Clare was supported in reflecting on how gender stereotypes marginalized them by seeing similarities between their story and the narrator's story, as elicited by the blocks of code in the code poetry interaction.

Mementorium provides a new way to interact with stories and experiences of marginalization, which situate the player as both the person actively recalling the narrator's past harm and as a separate person with their own experiences and knowledge who must decide how the narrator should react to these experiences. Through our design of the application and analysis of player interactions, we show how VR simulations do not need to insert players into a traumatic experience and sidestep critical learning processes of listening to marginalized people. Instead, we show how interactions can be designed to support players to stay emotionally connected to the narrator and reflect upon their experiences.

We have received feedback from the gaming and computer graphics industry that they would like to see cutscenes or vignettes. In our future development, we are looking to include more visuals during the narrative in the form of animation and to study the comparative impact this has on factors such as engagement, listening, and the player's ability to reflect. While this may improve the player's engagement, it is possible that visual images could be detrimental to listening and self-reflection because it externalizes the story and could prevent players from drawing upon their own memories to make connections with the narrator.

Next, our team plans to move Mementorium to the OpenXR framework so that it can be available on multiple VR platforms (cross-compatibility is currently only possible using Revive). Mementorium is one application along a continuum of applications that we have developed to explore interactive and immersive learning environments for learning about gender and sexuality-based marginalization [4, 6–8]. With each application that we develop and research, we are mapping the affordances of interactive and immersive technologies while developing insights into how to support learning about gender and sexuality as complex experiences with individual and systemic dimensions.

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